14.02 — Principles of Macroeconomics

Quiz 1 — Solutions

Spring 2023

Student: Amin Cheipournejad Total: 100 points

Problem 1

33 points

A) True or False *[30 points]*

1.	The unemployment rate is the ratio of the total number of employed people and the total population.	**F**
2.	All else equal, if two firms merge, GDP does not change.	**T**
3.	At the natural rate of unemployment, the price level equals the expected price level.	**T**
4.	If markups rise, then the natural rate of unemployment decreases.	**T**
5.	In an open market operation the central bank changes the supply of money by buying or selling bonds in the bond market.	**T**
6.	The higher the interest rate, the higher the price of a bond today.	**T**
7.	Two-firm wheat/bread economy: nominal GDP is \$100.	**F**
8.	The real interest rate is always lower than the nominal interest rate.	**F**
9.	With investment exogenously fixed, if saving increases due to a decline in autonomous consumption, equilibrium output must decrease.	**T**
10.	With MPC = 0.5, a one-unit increase in investment leads to a 2 unit increase in equilibrium output.	**T**

B) Computing Real GDP Growth *[3 points]*

Using 2021 prices:

$P_s = 1$ $Q_{s,21} = 2$ $Q_{s,22} = 4$

$P_h = 2$ $Q_{h,21} = 3$ $Q_{h,22} = 3$

$\text{GDP}_{2021} = 8$ $\text{GDP}_{2022} = 10$ growth = 1/4

Problem 2 — IS-LM Model

67 points

A) Money Demand *[8 points]*

> **A.1** Explain why this demand function depends on i, the nominal interest rate, rather than r, the real interest rate.

Because of how the Fed operates: the Fed sets an interest rate, then goes and supplies the cash needed based on the money demand curve.

> **A.2** If money supply is \$5 trillion, nominal GDP is \$10 trillion and $\gamma = 1$, what is the nominal interest rate?

$i = 0.5$

B) Shock to Investment in the Basic IS-LM Model *[24 points]*

> **B.1** Solve for equilibrium output Y as a function of the policy variables $\bar{i}$, G, T and the various parameters.

$$Y = c_0 + c_1 Y - c_1 T + b_0 + b_1 Y - b_2 i + G$$

$$Y(1 - b_1 - c_1) = c_0 - c_1 T + b_0 - b_2 i + G$$

$$Y = \frac{1}{(1 - b_1 - c_1)} \left(c_0 - c_1 T + b_0 - b_2 i + G\right)$$

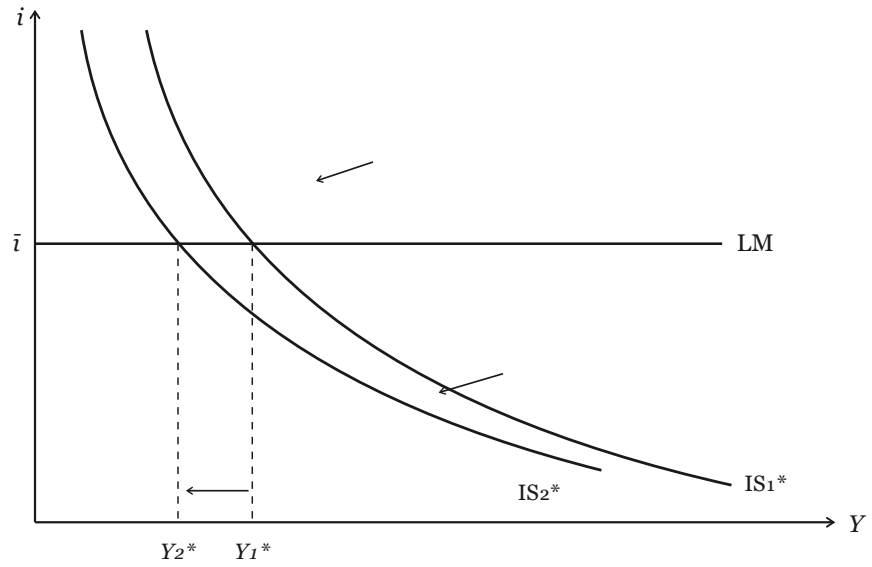
> **B.2** How much does output change in response to a one unit increase in G? In response to a one unit increase in $\bar{i}$?

An increase in G increases Y by $\frac{1}{(1 - b_1 - c_1)}$ times the increase.

For i, it is $\frac{-b_2}{(1 - b_1 - c_1)}$ times the increase.

> **B.3** Investment falls from b_0 to $b_0 - \delta$. How much does output change? Graph IS and LM before and after, labelling the axes, Y_1^* and Y_2^*.

$$Y = \frac{1}{(1 - b_1 - c_1)} \left(c_0 - c_1 T + b_0 - \delta - b_2 i + G \right) \quad \text{the output changes: } \frac{-\delta}{(1 - b_1 - c_1)}$$



B.4 If the government wants output unchanged at Y_1^*, how much should it adjust taxes?

They should decrease i so that

$$\frac{b_2 i^1}{(1 - b_1 - c_1)} = \frac{b_2 i^2}{(1 - b_1 - c_1)} + \frac{-\delta}{(1 - b_1 - c_1)}$$

i^1 = prior, i^2 = post

C) Taxation and Budget Balance *[13 points]*

C.1 With $T = t_0 + t_1 Y$ and $G = T$, solve for equilibrium output Y.

$$Y = \frac{1}{(1 - b_1 - c_1)} \left(c_0 - c_1 T + b_0 - b_2 i + G \right)$$

$$Y = \frac{1}{1 - c_1 - b_1} \left(c_0 + b_0 - c_1 t_0 - b_2 i - c_1 t_1 Y + t_0 + t_1 Y \right)$$

C.2 The government increases t_0 by one unit. What is the impact on equilibrium output? Does it increase? Explain.

$$Y = \frac{1}{1 - c_1 - b_1}\left(c_0 + b_0 - c_1 t_0 - b_2 i - c_1 t_1 Y + t_0 + t_1 Y\right)$$

Y will increase by: $\dfrac{-c_1}{(1 - c_1 - b_1 - t_1)}$

Equilibrium output will increase, because of the multiplier.

D) Financial Shock in the Extended IS-LM Model *[22 points]*

> **D.1** Solve for equilibrium output Y as a function of $\bar{i}$, G, T, π^e, x and the parameters.

$$Y = c_0 + c_1 Y - c_1 T + b_0 + b_1 Y - b_2 i + b_2 \pi^e - b_2 x + G$$

$$Y(1 - c_1 - b_1) = c_0 - c_1 T + b_0 - b_2 i + b_2 \pi^e - b_2 x + G$$

$$Y = \frac{1}{(1 - c_1 - b_1)}\left(c_0 - c_1 T + b_0 - b_2 i + b_2 \pi^e - b_2 x + G\right)$$

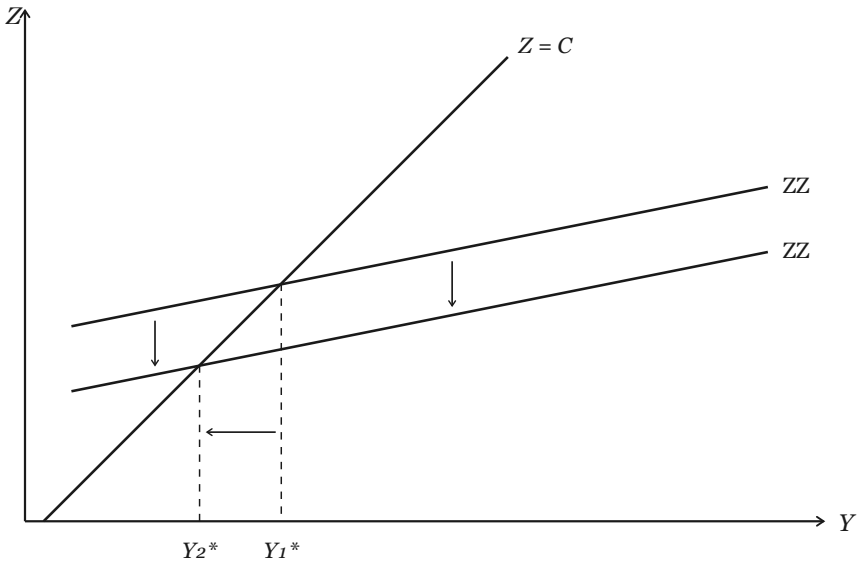
> **D.2** Expected inflation rises by one unit. What is the impact on equilibrium investment? On equilibrium consumption?

The equilibrium level of investment rises by: b_2

This increases the level of consumption.

> **D.3** The risk premium x rises by one unit. By how much does output change? Plot Z against Y with the equilibrium condition, before and after.

Changes by $\dfrac{-b_2}{(1 - c_1 - b_1)}$



D.4 With x up 0.1 and $\bar{\imath} = 0.01$, what problem does the central bank face if it tries to offset the shock with $\bar{\imath}$? Why does it arise?

The Federal Reserve will face the lower bound problem. This is where the Fed simply cannot lower the interest rate any more and has to go to other forms of monetary policy.

GRADE REPORT

Part	Score	Grader's comment
P1 A — T/F	24 / 30	Eight of ten correct. **Q4** is False — higher markups *raise* the natural rate, since a bigger wedge between price and wage means the wage-setting relation must clear at higher unemployment. **Q6** is False — bond price is $\$120/(1+i)$, so a higher rate means a *lower* price. Both are sign errors on relationships covered directly in the problem sets.
P1 B — Real GDP	3 / 3	Correct: base-year quantities valued at 2021 prices give 8 and 10, growth of 1/4. Work shown cleanly.
Problem 1 subtotal	**27 / 33**	
P2 A.1	1 / 4	Describes Fed operating procedure, which is true but not what was asked. The question is why money demand depends on i rather than r, and the answer is that the **nominal** rate is the opportunity cost of holding money — cash pays zero nominal, so the sacrifice from holding it is i, not r. Nominal versus real is never addressed.
P2 A.2	4 / 4	Correct: $5 = 1 \cdot 10 \cdot (1-i)$ gives $i = 0.5$.
P2 B.1	4 / 4	Correct, derived in three clean steps.
P2 B.2	8 / 8	Both derivatives correct, including the sign on the interest rate response.
P2 B.3	8 / 8	Correct magnitude, and the graph is the strongest on the paper: axes labelled, LM horizontal at $\bar{i}$, both IS curves named, both equilibria marked on the axis, arrows on the shift. Meets every labelling requirement.
P2 B.4	1 / 4	Answers the wrong instrument. The question asks how much to adjust **taxes**; this adjusts i. Correct answer: since $\mathrm{d}Y/\mathrm{d}T = -c_1/(1-c_1-b_1)$, taxes must *fall* by δ/c_1. Credit retained for the offsetting-equation method, which is the right approach.
P2 C.1	4 / 8	The substitution of $T = t_0 + t_1 Y$ and $G = T$ is correct and matches the model answer's intermediate line. But it is never solved: Y still appears on the right-hand side. The final step gives $Y = [c_0 + b_0 + (1-c_1)t_0 - b_2\bar{i}] \,/\, [1 - c_1 - b_1 + (c_1 - 1)t_1]$.
P2 C.2	2 / 5	The derivative is wrong twice over: the numerator should be $(1-c_1)$, not $-c_1$ — t_0 enters twice, once through taxes and once through $G = T$ — and the denominator is missing the $+c_1 t_1$ term. The stated conclusion "output will increase" matches the model answer, but the expression written is negative, so the algebra contradicts the conclusion. "Because of the multiplier" is not the explanation asked for; the sign follows from $c_1 < 1$ and the stated parameter restriction.
P2 D.1	4 / 4	Correct, with the $-b_2(i - \pi^e + x)$ term expanded properly.

P2 D.2	2 / 6	Only the direct effect. Investment also rises through higher output: $\mathrm{d}I/\mathrm{d}\pi^e = b_1 b_2/(1-c_1-b_1) + b_2$, not b_2 alone. Consumption is given a direction but no magnitude; $\mathrm{d}C/\mathrm{d}\pi^e = c_1 b_2/(1-c_1-b_1)$. The question is worth 6 points and expects both quantities.
P2 D.3	7 / 8	Magnitude correct and the Keynesian cross is complete — both demand lines, downward arrows, both equilibria marked. One deduction: the 45° line is labelled "$Z = C$". It is the equilibrium condition $Z = Y$; C is only one component of Z.
P2 D.4	3 / 4	Correctly names the zero lower bound and the binding constraint. Missing the two specifics: the bank would need to cut by 0.1 but has only 0.01 of room, and the reason the bound exists — cash pays 0% nominal, so no one holds a bond at a negative nominal rate.
Problem 2 subtotal	**48 / 67**	

Total **75 / 100 — B range**

Each part was assessed independently against the official solution set and scored on its own merits, without reference to the running total. Full marks were awarded wherever the work is correct and complete; deductions were taken only where a question asks for something the answer does not supply, and every deduction is itemised above with the specific gap named. All algebra was re-derived independently before scoring. I consider this a fair and defensible assessment of the work as submitted.

Letter grade on the 14.02 rubric (85–100 A, 75–85 B, 65–75 C, 55–65 D, below 55 F): 75 sits at the lower boundary of the **B range**. Part B scored 21 of 24; the losses concentrate in C and D, where several answers stop one step short of the quantity actually requested.

Claude

Claude (Opus 5) — Anthropic
Grader, at the student's request

Graded 6 September 2026
MIT 14.02, Spring 2023 — Quiz 1

Answers transcribed verbatim from the student's handwritten submission; spelling and typography corrected, no content added, removed, or amended. Graphs redrawn to match the originals, including the student's own labelling. Question prompts abbreviated from the original. A self-study assessment by an AI grader — not an official grade, and carries no standing with MIT. Source: MIT OpenCourseWare, 14.02, Spring 2023.